

1. A 25-year-old woman comes to the physician because of growths on her lips and tongue for several years. Examination shows nodules on the lips, buccal mucosa, and tongue; café au lait spots over the trunk and limbs; and scattered nodules on the skin. This patient is at greatest risk for which of the following conditions?

- ☐ A) Craniopharyngioma
- ☐ B) Ependymoma
- ☐ C) Lymphoma
- ☐ D) Optic nerve glioma
- ☐ E) Pituitary adenoma



Next

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Lab Values



Calculator



Review



Help



Pause

2. A 57-year-old woman comes to the physician because of a 3-month history of gradually progressive weakness in the shoulder and hip muscles. She has hypothyroidism treated with thyroxine but is otherwise healthy. Her vital signs are within normal limits. Neurologic examination shows moderate bilateral weakness of the deltoid, biceps, triceps, psoas, glutei, and quadriceps muscles. The remainder of the examination shows no abnormalities. Her erythrocyte sedimentation rate is 85 mm/h, and serum creatine kinase activity is 2456 U/L. Complete blood count and urinalysis are within normal limits. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Quantitative sensory testing
- ☐ B) Electromyography and nerve conduction studies
- ☐ C) MRI of the spinal cord
- ☐ D) Skin biopsy
- ☐ E) Lumbar puncture



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

3. A 27-year-old man is brought to the emergency department because of multiple trauma sustained in a motor vehicle collision. He is initially arousable, but he is increasingly unresponsive over the next 45 minutes. The pupil of the left eye is dilated, and there is ptosis of the left eyelid. He begins to hyperventilate. Which of the following is the most likely cause?

- | | |
|--|--|
| ☐ A) Alcohol withdrawal | ☐ G) HIV encephalopathy |
| ☐ B) Aseptic meningitis | ☐ H) Infratentorial lesion |
| ☐ C) Bacterial meningitis | ☐ I) Myxedema |
| ☐ D) Carbon monoxide poisoning | ☐ J) Salicylate intoxication |
| ☐ E) Central herniation | ☐ K) Uncal herniation |
| ☐ F) Factitious disorder | ☐ L) Viral meningitis |



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

4. One day after undergoing an elective cholecystectomy, a 67-year-old woman becomes confused and belligerent. She has a 2-year history of mild dementia, Alzheimer type. She takes no medications. Intramuscular haloperidol is administered, and 1 hour later, she begins to repetitively hyperextend her neck. She also has repetitive conjugate upward movements of the eyes that she says she cannot control. Examination shows no other abnormalities. Administration of a medication with which of the following properties is the most appropriate treatment?

- ☐ A) Anticholinergic
- ☐ B) Antidopaminergic
- ☐ C) Antiserotonergic
- ☐ D) Cholinergic
- ☐ E) Dopaminergic
- ☐ F) Serotonergic



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

5. A previously healthy 32-year-old woman comes to the physician because of a 1-week history of progressive sensory loss. Initially, only her feet felt numb, but the loss of sensation has slowly ascended symmetrically to the umbilicus. She also has had urinary urgency and frequency, nocturia, and the perception of a tight band-like sensation around her midabdominal region. Examination shows slowing of left eye adduction during saccadic movement of the eyes to the right. There is diffuse hyperreflexia. Sensation to pinprick is decreased to the level of the umbilicus. Which of the following is the most likely diagnosis?
- ☐ A) Adrenal leukodystrophy
 - ☐ B) Cytomegalovirus polyradiculopathy
 - ☐ C) Guillain-Barré syndrome
 - ☐ D) Multiple cerebral infarcts
 - ☐ E) Multiple sclerosis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

6. A 67-year-old woman is brought to the physician 3 hours after a 5-minute episode of visual loss in her left eye. She has not had any other symptoms. She has hypertension and stable angina pectoris. Medications include lisinopril and metoprolol. Her pulse is 72/min, and blood pressure is 150/92 mm Hg. Funduscopy examination shows no abnormalities. A grade 3/6 holosystolic murmur is heard best at the apex. Neurologic examination shows no focal findings. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Complete blood count
 - ☐ B) Measurement of erythrocyte sedimentation rate
 - ☐ C) Visual field testing
 - ☐ D) MRI of the brain
 - ☐ E) Carotid duplex ultrasonography
 - ☐ F) Ocular tonometry



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

7. An 11-year-old girl is brought to the emergency department by her mother because of a 2-day history of increasing weakness and a prickling sensation in her legs. She was previously healthy except for a mild flu-like illness during the past week. Her temperature is 37°C (98.6°F), pulse is 78/min, respirations are 20/min, and blood pressure is 115/70 mm Hg. Examination of the lower extremities shows a muscle strength of 1/5 and decreased sensation to pinprick. Achilles tendon and patellar reflexes are decreased. The remainder of the examination shows no abnormalities. A lumbar puncture is performed. Analysis of cerebrospinal fluid shows a total protein concentration of 80 mg/dL and a leukocyte count of 4/mm³. In addition to close observation, oxygen therapy, and maintenance of fluids, which of the following is the most appropriate pharmacotherapy?

- ☐ A) Antibiotics
- ☐ B) Anticonvulsants
- ☐ C) Antivirals
- ☐ D) Corticosteroids
- ☐ E) Immune globulins



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

8. A 42-year-old woman with alcoholism has severe burning paresthesias of both feet and decreased vibratory sensation in both ankles 2 months after beginning treatment for active, cavitary pulmonary tuberculosis. At the time of diagnosis she was malnourished, and isoniazid, rifampin, ethambutol, vitamin B₆, and folic acid were prescribed. The most likely cause of paresthesias of her feet and decreased sensation in her ankles is failure to take which of the following prescribed medications?

- ☐ A) Ethambutol
- ☐ B) Folic acid
- ☐ C) Isoniazid
- ☐ D) Rifampin
- ☐ E) Vitamin B₆



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

9. A 20-year-old college student comes to the physician for a follow-up examination 2 days after sustaining a head injury when he ran into another player while playing flag football. He did not lose consciousness but does not remember anything from the rest of the game. He was evaluated in the emergency department 1 hour after the injury; a CT scan of the head showed no abnormalities and he was discharged. Today, he reports a mild, persistent headache that improves with acetaminophen. He has not had nausea, vomiting, or memory problems and was able to return to school today. Examination shows no abnormalities. He asks about playing in a flag football playoff game tomorrow. Which of the following is the most appropriate recommendation?

- ☐ A) No contact sports pending evaluation in 1 week
- ☐ B) No contact sports pending an MRI of the brain
- ☐ C) No contact sports pending a lumbar puncture
- ☐ D) No contact sports for 1 year; schedule regular follow-up examinations
- ☐ E) Return to sports; no further management is indicated at this time



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

10. A 19-year-old woman is brought to the emergency department by paramedics after having a seizure at a dormitory party on campus. According to her boyfriend, who accompanied her, she had been drinking more water than usual and then became silent and began staring. He then witnessed her collapse to the floor and begin shaking for approximately 1 minute. On arrival, she is mute and unresponsive to questions. She appears restless and periodically grinds her teeth. Her temperature is 39.6°C (103.3°F), pulse is 90/min, respirations are 24/min, and blood pressure is 150/95 mm Hg. Physical examination shows profuse diaphoresis and piloerection. Pupils are 5 mm bilaterally. The remainder of the examination shows no abnormalities. Laboratory studies show:

Prothrombin time	16 sec
Serum	
Na ⁺	114 mEq/L
Urea nitrogen	40 mg/dL
Creatinine	2 mg/dL
ALT	90 U/L
Creatine kinase	1200 U/L

The most likely diagnosis is intoxication with which of the following substances?

- ☐ A) Amitriptyline
- ☐ B) Clonazepam
- ☐ C) Ecstasy (3,4-methylenedioxymethamphetamine)
- ☐ D) γ-Hydroxybutyrate
- ☐ E) Phencyclidine



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

11. A 47-year-old man with a history of alcoholism is brought to the emergency department by his family because of confusion. He lives alone, and they found him disoriented 4 hours ago. An empty plastic container with a sweet odor was found in his apartment. He is oriented only to place. Cranial nerves, motor strength, and sensation are normal. Serum studies show a creatinine concentration of 1.5 mg/dL, bicarbonate concentration of 8 mEq/L, and an increased anion gap. Serum ketones test is negative. Urinalysis shows crystals. Exposure to which of the following is the most likely cause of these findings?

- ☐ A) Acetaminophen
- ☐ B) Arsenic
- ☐ C) Carbon monoxide
- ☐ D) Cyanide
- ☐ E) Ethanol
- ☐ F) Ethylene glycol
- ☐ G) Methanol
- ☐ H) Organophosphate
- ☐ I) Salicylates



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

12. A 77-year-old man with hypertension comes to the physician for a routine follow-up examination. He feels well. His only medication is a diuretic. His blood pressure is 140/70 mm Hg. On neurologic examination, cranial nerves are intact. Muscle tone and strength are normal. Deep tendon reflexes are absent in the ankles but otherwise normal. Which of the following is the most appropriate next step in management?

- ☐ A) Measurement of urine lead concentration
- ☐ B) MRI of the lumbar spine
- ☐ C) Electromyography and nerve conduction studies
- ☐ D) Lumbar puncture
- ☐ E) No additional management is indicated



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

13. An otherwise healthy 27-year-old man comes to the physician because of a 3-month history of decreased libido and delayed ejaculation. He says that his symptoms are having a detrimental effect on his relationship. He has not had similar problems in the past. Four months ago, he was evaluated for low mood and loss of interest and productivity; his symptoms have been well controlled with sertraline since that time. He takes no other medications. Physical examination shows no abnormalities. Which of the following neurotransmitters is most likely responsible for this patient's current symptoms?

- ☐ A) Acetylcholine
- ☐ B) γ -Aminobutyric acid
- ☐ C) Dopamine
- ☐ D) Norepinephrine
- ☐ E) Serotonin



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

14. A previously healthy 27-year-old man comes to the physician because of a 2-week history of pain in his neck and left shoulder radiating through the lateral surface of his left arm to his thumb and index finger. Examination shows a decreased biceps reflex on the left. Muscle strength is 4/5 on extension of the left wrist. Sensation to light touch is decreased over the left thumb and index finger. The remainder of the examination, including neurologic examination, shows no abnormalities. Which of the following is the most likely diagnosis?
- ☐ A) C4 radiculopathy
 - ☐ B) C5 radiculopathy
 - ☐ C) C6 radiculopathy
 - ☐ D) C7 radiculopathy
 - ☐ E) C8 radiculopathy
 - ☐ F) Carpal tunnel syndrome
 - ☐ G) Cubital tunnel syndrome



15. A 72-year-old man comes to the physician because of a 7-day history of increasingly severe pain around the waist. He says that it feels like a burning belt from the left midback to the umbilicus. He has a history of chronic low back pain. He has type 2 diabetes mellitus treated with glyburide and asthma treated with beclomethasone and albuterol inhalers. He completed a 10-day course of corticosteroids for asthma 3 weeks ago. Examination shows numbness in the left T12 distribution. The remainder of the examination shows no abnormalities. Which of the following is the most likely explanation for these findings?
- ☐ A) Ankylosing spondylitis
 - ☐ B) Atherosclerotic vascular disease
 - ☐ C) Diabetic radiculopathy
 - ☐ D) Epidural abscess
 - ☐ E) Low back strain
 - ☐ F) Lumbar radiculopathy
 - ☐ G) Lumbar stenosis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

16. A 12-year-old girl with systemic lupus erythematosus is brought to the emergency department by her parents after they had difficulty awakening her this morning. She has needed constant stimulation to stay awake. Over the past 2 days, she has had a flu-like illness and vomiting. She has been unable to take her corticosteroid and antihypertensive medications. Her temperature is 37°C (98.6°F), pulse is 60/min, respirations are 40/min, and blood pressure is 180/130 mm Hg. She is obtunded but moves all extremities to painful stimuli. Funduscopic examination shows bilateral papilledema with flame hemorrhages. Deep tendon reflexes are brisk and symmetric. Her erythrocyte sedimentation rate is 12 mm/h. Which of the following is the most likely diagnosis?

- ☐ A) Brain neoplasm
- ☐ B) Cerebral vasculitis
- ☐ C) Hypertensive encephalopathy
- ☐ D) Idiopathic intracranial hypertension
- ☐ E) Intracerebral hematoma
- ☐ F) Reye syndrome



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

17. A previously healthy 4-month-old girl is brought to the emergency department by her parents because of a 3-day history of increased drowsiness. During this period, she also has been feeding poorly and "spitting up" after each feeding. She has not had fever. She was born at term following an uncomplicated pregnancy and delivery. On arrival, she is responsive to tactile stimuli. She is at the 25th percentile for length and weight and 90th percentile for head circumference. Her temperature is 37°C (98.6°F), pulse is 80/min, respirations are 24/min, and blood pressure is 100/70 mm Hg. Examination shows a bulging anterior fontanel. There is a hyperpigmented macule with hair growth on the upper lumbar spine; her parents report that this finding was noted at birth. Which of the following is the most likely diagnosis?

- ☐ A) Bacterial meningitis
- ☐ B) Chiari type 2 malformation
- ☐ C) Idiopathic increased intracranial pressure
- ☐ D) Intracranial schwannoma
- ☐ E) Tuberous sclerosis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

18. A 13-year-old girl is brought to the physician because of headaches accompanied by nausea and vomiting over the past 2 weeks; they are located behind the left eye and improve after sleep. The headaches started during final examinations, and she now is missing 2 to 3 days of school weekly because of the pain. Her temperature is 37°C (98.6°F), pulse is 90/min, respirations are 20/min, and blood pressure is 95/60 mm Hg. Examination shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Brain tumor
- ☐ B) Cluster headaches
- ☐ C) Migraine
- ☐ D) Sinusitis
- ☐ E) Tension-type headaches



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

19. Over the past year, a 72-year-old man has had progressive aching in his lower extremities when he walks in the shopping mall. Initially, the pain began after 15 minutes of walking and resolved after he stopped; now, it starts after 5 or 10 minutes of walking, and it occasionally occurs when he stands up after sleeping all night. The pain radiates from his buttocks to his feet. He describes it as a "terrible" pain. He has mild chronic obstructive pulmonary disease but continues to smoke one pack of cigarettes daily. He has cervical arthritis and often has neck and arm pain. Examination shows decreased breath sounds bilaterally. Strength and sensation are normal in the extremities. Babinski's sign is absent bilaterally. Deep tendon reflexes are 1+ in both knees and ankles and 2+ in the upper extremities. Dorsalis pedis pulses are normal. Which of the following is the most likely diagnosis?

- | | |
|--|---|
| ☐ A) Alcoholic rhabdomyolysis | ☐ J) Hypokalemic periodic paralysis |
| ☐ B) Amyotrophic lateral sclerosis | ☐ K) Hypothyroid myopathy |
| ☐ C) Becker muscular dystrophy | ☐ L) Lumbar spinal stenosis |
| ☐ D) Botulism | ☐ M) Multiple sclerosis |
| ☐ E) Carcinomatous meningitis | ☐ N) Myasthenia gravis |
| ☐ F) Cervical spondylosis | ☐ O) Polymyositis |
| ☐ G) Epidural metastatic carcinoma | ☐ P) Sensorimotor peripheral neuropathy |
| ☐ H) Epstein-Barr virus infection | ☐ Q) Spinal cord compression |
| ☐ I) Guillain-Barré syndrome | |



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

20. A 25-year-old man comes to the physician because of a 4-month history of increasing low back pain that is most severe on awakening in the morning. His pain is relieved by stretching. There is no history of trauma. He is otherwise healthy except for occasional dysuria. Examination shows decreased forward flexion of the spine and tenderness over the lumbosacral area and at the insertion site of the Achilles tendon. Which of the following is the most likely diagnosis?

- | | |
|--|---|
| ☐ A) Abdominal aortic aneurysm | ☐ G) Metastatic cancer |
| ☐ B) Ankylosing spondylitis | ☐ H) Osteoporosis |
| ☐ C) Herniated disc at L4–L5 | ☐ I) Pyelonephritis |
| ☐ D) Intervertebral discitis | ☐ J) Rheumatoid arthritis |
| ☐ E) Ischemia | ☐ K) Spondylolisthesis |
| ☐ F) Lumbar spinal stenosis | ☐ L) Systemic lupus erythematosus |



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

21. A 67-year-old woman is brought to the emergency department 1 hour after a 25-minute episode of right facial droop, weakness of the right arm and hand, and an inability to talk. She has hypertension, type 2 diabetes mellitus, coronary artery disease, chronic atrial fibrillation, and gout. Her medications include metoprolol, fosinopril, glyburide, and allopurinol. Her pulse is 80/min and irregularly irregular, respirations are 18/min, and blood pressure is 170/100 mm Hg. Cardiac examination shows no murmurs. Sensation is decreased to pinprick in the distal lower extremities. Achilles tendon reflexes are decreased. The remainder of the neurologic examination shows no abnormalities. A CT scan of the head shows no areas of infarction or hemorrhage. Echocardiography shows dilation of the left atrium with thrombus in the left atrial appendage, left ventricular hypertrophy, and mild mitral regurgitation. Carotid ultrasonography shows mild stenosis of both the right and left internal carotid arteries without thrombus or ulceration. Which of the following is the most appropriate pharmacotherapy to prevent similar episodes in the future?

- ☐ A) Aspirin
- ☐ B) Clopidogrel
- ☐ C) Dipyridamole
- ☐ D) Ticlopidine
- ☐ E) Warfarin



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

The response options for the next **2** items are the same. Select one answer for each item in the set.

For each patient who has transient episodes of acute neurologic dysfunction, select the most likely diagnosis.

- | | |
|--|--|
| ☐ A) Cataplexy | ☐ G) Simple partial (motor) seizure with generalization |
| ☐ B) Complex partial seizure | ☐ H) Simple partial (motor) seizure without generalization |
| ☐ C) Migraine | ☐ I) Simple partial (sensory) seizure with generalization |
| ☐ D) Panic attack | ☐ J) Simple partial (sensory) seizure without generalization |
| ☐ E) Photomyoclonic seizure | ☐ K) Syncope |
| ☐ F) Primary generalized seizure | ☐ L) Transient ischemic attack |

22. An otherwise healthy 52-year-old man has episodes of loss of consciousness preceded by twitching of the right thumb, then the hand, the wrist, the elbow, and the shoulder. The spread from thumb to shoulder takes about 20 seconds. He then finds himself on the floor; he often bites his tongue and urinates during these episodes.



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

For each patient who has transient episodes of acute neurologic dysfunction, select the most likely diagnosis.

- | | |
|--|--|
| ☐ A) Cataplexy | ☐ G) Simple partial (motor) seizure with generalization |
| ☐ B) Complex partial seizure | ☐ H) Simple partial (motor) seizure without generalization |
| ☐ C) Migraine | ☐ I) Simple partial (sensory) seizure with generalization |
| ☐ D) Panic attack | ☐ J) Simple partial (sensory) seizure without generalization |
| ☐ E) Photomyoclonic seizure | ☐ K) Syncope |
| ☐ F) Primary generalized seizure | ☐ L) Transient ischemic attack |

23. A 72-year-old man has had episodes of loss of consciousness for the past 2 years. He says that there are no warning symptoms and that he suddenly finds himself on the ground. He has fractured his arm and has bruised himself many times. Bystanders say that he suddenly becomes pale and sweaty and then slumps to the floor like a rag doll. Jerking or twitching movements of the trunk and extremities occur for several seconds during episodes. He recovers completely and is not confused at the end of the episode. He has had two myocardial infarctions.



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

24. A 67-year-old man is brought to the physician by his wife because of a 6-week history of progressive cognitive changes and "twitching movements" of his arms and legs. He has no history of serious illness and takes no medications. Vital signs are within normal limits. Examination shows mild ataxic dysarthria. Muscle strength is 5/5 throughout. Deep tendon reflexes are 3+ throughout. He has an ataxic gait and frequent irregular jerks of the upper and lower extremities. On mental status examination, he is alert but distractible. His Mini-Mental State Examination score is 19/30. A lumbar puncture is performed. Cerebrospinal fluid (CSF) analysis shows:

Glucose	80 mg/dL
Total protein	45 mg/dL
WBC	none
RBC	none

An MRI of the brain shows vague areas of abnormal T2 signal in the basal ganglia; these areas have no mass effect and do not enhance. Additional CSF analysis is most likely to show which of the following abnormalities?

- ☐ A) Abnormal cytologic study results
- ☐ B) Increased 14-3-3 protein concentration
- ☐ C) Positive herpes simplex polymerase chain reaction test
- ☐ D) Positive serologic test for syphilis
- ☐ E) Presence of anti-Hu antibody



25. A 47-year-old woman comes to the physician because of a 6-month history of gradual hearing loss in the left ear. She also has had difficulty maintaining her balance over the past 3 months. Six months ago, she had an episode of nephrolithiasis complicated by a severe urinary tract infection requiring intravenous antibiotic therapy. Examination shows a decrease of the left nasolabial fold and a slight perioral droop when she smiles. The forehead movements are intact bilaterally. Her gait is normal, but she has difficulty with tandem walking. Audiometry shows moderate high-frequency hearing loss on the left with impaired speech discrimination. Which of the following is the most likely diagnosis?

- ☐ A) Acoustic neuroma (vestibular schwannoma)
- ☐ B) Aminoglycoside ototoxicity
- ☐ C) Brain stem infarction
- ☐ D) Mastoiditis
- ☐ E) Meniere disease



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

26. A previously healthy 47-year-old man comes to the physician because of an 8-month history of progressively severe bifrontal headache. He has a scant beard and a sallow, waxy complexion. Examination shows bilateral temporal hemianopia. Visual acuity is normal. Which of the following is the most likely diagnosis?

- | | |
|--|---|
| ☐ A) Cluster headaches | ☐ G) Pituitary adenoma |
| ☐ B) Craniopharyngioma | ☐ H) Subdural hematoma |
| ☐ C) Medulloblastoma | ☐ I) Temporal arteritis |
| ☐ D) Migraine | ☐ J) Temporal lobe glioma |
| ☐ E) Occipital neuralgia | ☐ K) Temporomandibular joint syndrome |
| ☐ F) Parasagittal meningioma | |



27. A 25-year-old woman comes to the physician because of a 6-month history of worsening excessive daytime sleepiness, even during important events such as meetings at work. She says she has trouble sustaining her attention and that her symptoms are beginning to impair her work productivity. She also reports occasional episodes in which her muscles go limp and she falls suddenly. She has injured herself during two of these episodes. She is 163 cm (5 ft 4 in) tall and weighs 59 kg (130 lb); BMI is 22 kg/m². Her pulse is 85/min, respirations are 12/min, and blood pressure is 120/76 mm Hg. Physical examination shows no abnormalities. Which of the following is the most appropriate initial step in management?

- ☐ A) Weight loss program
- ☐ B) Continuous positive airway pressure
- ☐ C) Phototherapy
- ☐ D) Clonazepam therapy
- ☐ E) Modafinil therapy
- ☐ F) Uvulopalatopharyngoplasty



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

28. A 68-year-old woman comes to the physician because of a 3-month history of a spinning sensation. She reports feeling as though the room is spinning particularly when she tries to turn over in bed; the feeling usually resolves within 1 minute. She has no vomiting with an episode but does feel a little queasy. Examination shows normal hearing. Ocular movements are full when the patient is sitting, but when her head is allowed to hang off the examination table, she develops a coarse, rotatory nystagmus beating to one side. The remainder of the examination shows no abnormalities. The most likely cause of this patient's condition is an abnormality at which of the following sites?

- ☐ A) Auditory cortex
- ☐ B) Brain stem
- ☐ C) Otolith
- ☐ D) Ossicle
- ☐ E) Spiral ganglion



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

29. A 10-year-old girl is brought to the physician by her parents after the Thanksgiving holiday break because of a 4-day history of rapid, irregular involuntary movements of her face, trunk, and extremities. Her parents say that during this period, she has been anxious and irritable and has had headaches. She seems to be worrying obsessively about things that formerly would not have worried her. She has no history of serious illness except for migratory joint pain that occurred in the early fall. Examination shows restless-appearing, generalized involuntary movements of the face, trunk, and extremities. Which of the following is the most likely cause of this patient's symptoms?

- ☐ A) Autoimmune disorder of the basal ganglia
- ☐ B) Conversion disorder
- ☐ C) Heavy metal toxicity
- ☐ D) Hypercalcemia
- ☐ E) Hypothyroidism
- ☐ F) Tandem repeat mutation on chromosome 4



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

30. A 32-year-old woman with breast cancer metastatic to bone is brought to the emergency department because of a 12-hour history of severe pelvic pain and numbness. She has not urinated for more than 24 hours. Her only medication is hydrocodone. On arrival, her temperature is 37.2°C (99°F), pulse is 104/min, respirations are 18/min, and blood pressure is 152/70 mm Hg. Rectal examination shows decreased rectal sphincter tone. Muscle strength is 4/5 in the lower extremities. Sensation to light touch is decreased in a saddle distribution. Abdominal examination shows suprapubic fullness. Placement of a urinary catheter yields 800 mL of clear yellow urine and resolves the pelvic pain but not the numbness. An MRI of the spine shows a tumor pushing into the L4–S1 intrathecal space. Which of the following is the most appropriate immediate next step in management?

- ☐ A) Oral administration of high-dose ibuprofen
- ☐ B) Intravenous administration of high-dose dexamethasone
- ☐ C) Intravenous administration of chemotherapy
- ☐ D) Intrathecal administration of chemotherapy
- ☐ E) Radiation therapy to the spine



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

31. A 42-year-old woman is brought to the emergency department by her husband after he found her in a confused state when he returned home from a 3-day fishing trip. She has a long-standing relapsing psychiatric condition controlled with lithium carbonate, haloperidol, and clonazepam. She does not drink alcohol or use illicit drugs. She appears drowsy. Her temperature is 37.5°C (99.5°F), pulse is 85/min, respirations are 16/min, and blood pressure is 140/90 mm Hg. The pupils are 2 mm and reactive. There is a coarse tremor of the upper extremities. Deep tendon reflexes are increased throughout. She responds only to vigorous stimulation. On mental status examination, her speech is slow and slurred. She appears perplexed and sleepy and does not respond to questions. Laboratory studies show:

Leukocyte count	14,000/mm ³
Serum	
Urea nitrogen	60 mg/dL
Creatinine	2.5 mg/dL
Lithium	3.5 mEq/dL (therapeutic range=0.6–1.2)

In addition to discontinuing lithium carbonate, which of the following is the most appropriate next step in management?

- ☐ A) Observation only
- ☐ B) Discontinue haloperidol
- ☐ C) Add intravenous dantrolene
- ☐ D) Add intravenous flumazenil
- ☐ E) Begin hemodialysis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

32. A previously healthy 62-year-old man comes to the physician because of a 3-week history of intermittent drooping of his left eyelid and episodes of double vision. Neurologic examination shows bilateral ptosis, more prominent on the left. There is weakness of left eye abduction and adduction and weakness of right eye abduction. Pupillary reflexes are normal. The remainder of the examination shows no abnormalities. This patient is at greatest risk for which of the following?

- ☐ A) Colon cancer
- ☐ B) Lymphoma
- ☐ C) Prostate cancer
- ☐ D) Small cell lung carcinoma
- ☐ E) Thymoma



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



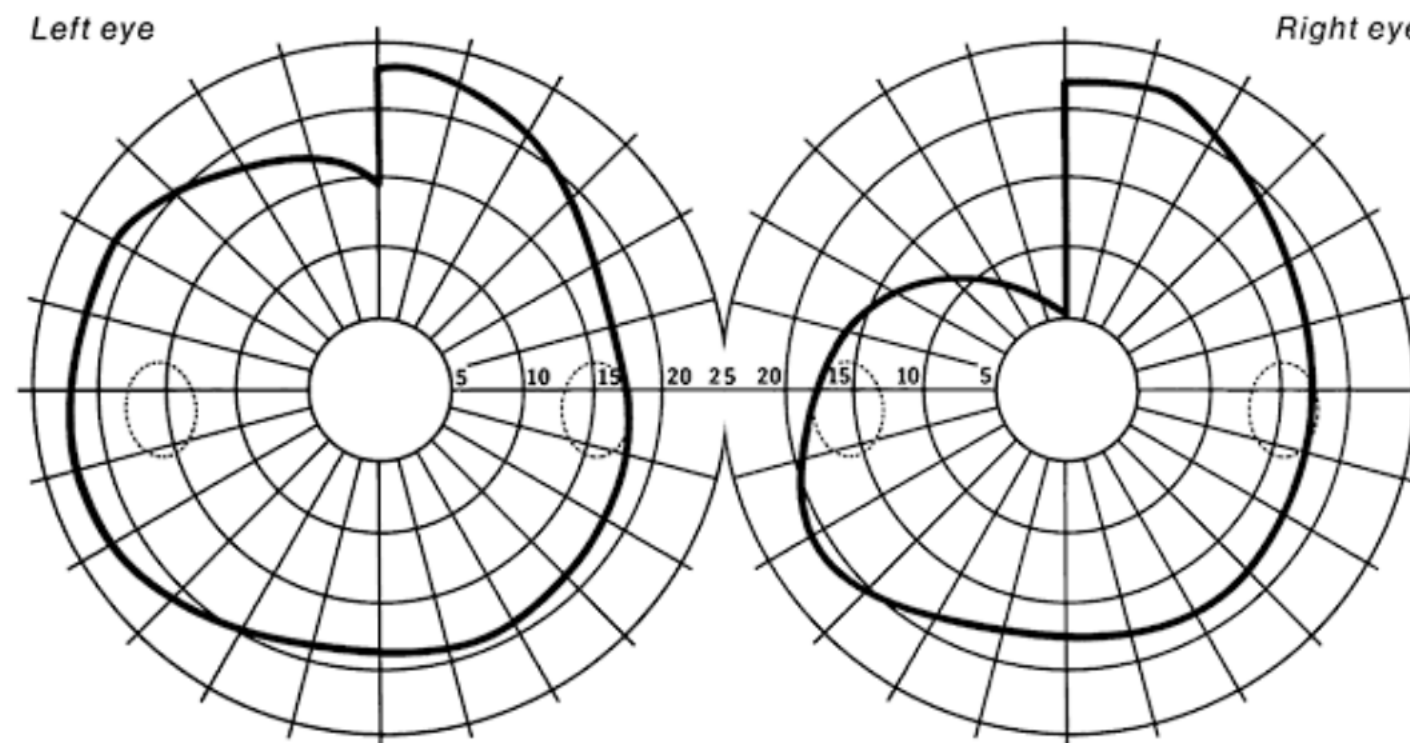
Review



Help



Pause



33. Over the past 3 months, a 60-year-old woman has had 2-minute episodes characterized by a rising sensation in the abdomen followed by loss of awareness. During an episode, she stares, makes chewing movements, and makes fumbling movements with her hands; afterwards, she is confused for a few minutes. Visual fields are shown. Neurologic examination shows no other abnormalities. The most likely diagnosis is a lesion at which of the following locations?

- ☐ A) Frontal lobe
- ☐ B) Occipital lobe
- ☐ C) Optic nerve
- ☐ D) Parietal lobe
- ☐ E) Temporal lobe



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

34. A previously healthy 57-year-old man comes for a follow-up examination because of progressive weakness of his arms and legs. Six months ago, examination showed no abnormalities except for 4/5 muscle strength in all extremities. Current examination shows fasciculations in the hands and right lower extremity and pseudobulbar palsy. Sensation is normal and cranial nerves are intact. He is oriented, and affect is appropriate. An MRI of the brain shows no abnormalities. Which of the following is the most likely mechanism of this patient's disease?
- ☐ A) Amyloid deposition
 - ☐ B) Demyelination
 - ☐ C) Motor neuron degeneration
 - ☐ D) Neoplastic infiltration of the central nervous system
 - ☐ E) Neuronal ganglioside GM₂ deposition
 - ☐ F) Neuronal loss in the substantia nigra



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

35. A 62-year-old man is brought to the physician by his wife because of a 3-year history of progressive forgetfulness. He has difficulty remembering names, especially of people he has recently met. He retired from his profession as a retail clerk 12 weeks ago because he had been forgetting the names of customers and new products and could no longer effectively perform his job. He also has had progressive difficulty with activities of daily living; his wife now chooses his clothes for him. He has no history of head trauma. He has no history of serious illness and takes no medications. He does not drink alcohol or use illicit drugs. Physical examination shows no abnormalities. On mental status examination, he has an indifferent mood and affect. He is oriented to person, place, and time but takes a few minutes before he correctly names the date. His Mini-Mental State Examination score is 22/30. There is no evidence of auditory or visual hallucinations. Results of laboratory studies are within the reference ranges. Administration of a drug from which of the following classes is most appropriate for this patient?

- ☐ A) Acetylcholinesterase inhibitor
- ☐ B) Dopamine agonist
- ☐ C) NMDA receptor agonist
- ☐ D) NMDA receptor antagonist
- ☐ E) Selective serotonin reuptake inhibitor



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

36. A 32-year-old man is brought to the emergency department by ambulance 20 minutes after he was found by a janitor on the floor of a restroom at a bus station. At the scene, he was having continuous, generalized tonic-clonic seizures. An endotracheal tube was placed, and supplemental oxygen was administered. On arrival to the emergency department, he is flushed and diaphoretic. There are no external signs of trauma. His temperature is 37.2°C (99°F), pulse is 110/min, respirations are 15/min, and blood pressure is 140/70 mm Hg. The pupils are 4 mm and reactive to light. Examination shows tonic stiffening of the trunk and synchronous clonic jerking of the extremities. In addition to intravenous administration of dextrose, which of the following is the most appropriate next step in treatment?

- ☐ A) Fosphenytoin
- ☐ B) Lorazepam
- ☐ C) Naloxone
- ☐ D) Phenobarbital
- ☐ E) Valproic acid



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

37. A previously healthy 37-year-old man is brought to the emergency department because of a 10-day history of increasingly severe headaches. The pain has been unremitting for the past 18 hours; he describes it as severe and pounding. His pulse is 70/min, respirations are 16/min, and blood pressure is 142/90 mm Hg. Examination of the face and neck shows mild swelling and a dusky reddish-purple skin color with venous distention over the head and neck; the skin color below the neck is normal. Examination of the optic fundi shows papilledema. Neurologic examination shows no focal findings. Which of the following is the most likely cause of this patient's increased intracranial pressure?

- ☐ A) Dilation of cerebral arteries
- ☐ B) Intracranial mass lesion
- ☐ C) Obstruction of cerebrospinal fluid flow
- ☐ D) Obstruction of venous return to the heart
- ☐ E) Overproduction of cerebrospinal fluid



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

38. Five days after a left hemicolectomy for colon cancer, a 62-year-old man is transferred to the intensive care unit because of temperatures to 40.6°C (105°F), lethargy, diaphoresis, and labile blood pressure for 3 hours. His postoperative course had been complicated by temperatures to 37.8°C (100°F) and nocturnal agitation since day 2; he has required haloperidol therapy (2 mg nightly). He has a history of bipolar disorder. Medications include morphine by patient-controlled analgesia pump, lithium carbonate, and clonazepam. He is lying expressionless in a pool of sweat and is drooling. His temperature is 40.6°C (105°F), pulse is 140/min, and blood pressure is 180/100 mm Hg. Examination shows wet skin and dry mucosa. There is a herpetic lesion on his lower lip. The neck is supple and the intravenous catheter insertion site is nontender with no erythema. Bowel sounds are hyperactive; there is no organomegaly. The wound is clean and dry with no erythema. Neurologic examination shows a masked facies and a marked intention tremor. Deep tendon reflexes are hyperactive and symmetric. Babinski sign is absent. Laboratory studies show:

Hemoglobin	15 g/dL
Leukocyte count	20,000/mm ³
Segmented neutrophils	80%
Bands	5%
Lymphocytes	10%
Monocytes	5%
Serum	
Urea nitrogen	38 mg/dL
Creatinine	1.6 mg/dL
Creatine kinase	1000 U/L

Urinalysis and an x-ray of the chest show no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Antibiotic-related fever
- ☐ B) Herpes encephalitis
- ☐ C) Lithium toxicity
- ☐ D) Malignant hyperthermia
- ☐ E) Neuroleptic malignant syndrome
- ☐ F) Postoperative sepsis



39. A 32-year-old woman is brought to the emergency department because of progressive visual dimming in her right eye since awakening 8 hours ago. Her symptoms began with a mild headache over her right eye before going to bed last night. She says this headache is different than her usual migraines that have improved since she started treatment with propranolol. Funduscopy examination shows no abnormalities of the right eye, but the patient cannot read and can barely count fingers with the right eye. Visual acuity in the left eye is normal. A CT scan of the head shows normal findings. Which of the following is the most likely diagnosis?

- ☐ A) Diabetic retinopathy
- ☐ B) Fabry disease
- ☐ C) Migraine
- ☐ D) Ophthalmic arteritis
- ☐ E) Optic neuritis
- ☐ F) Retinal detachment
- ☐ G) Retinal emboli
- ☐ H) Retinitis pigmentosa



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

40. A previously healthy 47-year-old man is brought to the physician because of a 2-week history of progressive weakness. He says that during this period, his feet "flop" while walking, and he trips when walking up stairs. During the past week, he has had difficulty standing up from low chairs and opening jars. He has not had pain or tingling. He has had constipation but no changes in bladder or sexual function. He has not had any recent infections or vaccinations. He takes no medications. He has smoked one pack of cigarettes daily for 20 years. He drinks three to four beers daily; for the past 6 weeks he has been drinking home-distilled whiskey. Examination shows atrophy of the intrinsic muscles of the hands and feet. Muscle strength is 4/5 in the wrist and finger extensors, interosseous, iliopsoas, and hamstring muscles and 3/5 in the anterior tibialis, peronei, and posterior tibialis muscles. Deep tendon reflexes are 1+ diffusely. Babinski sign is absent. Sensory examination shows no abnormalities. Laboratory studies show microcytic anemia and hyperuricemia. Which of the following is the most likely cause of this patient's neurologic symptoms?

- ☐ A) Acute intermittent porphyria
- ☐ B) Alcohol abuse
- ☐ C) Guillain-Barré syndrome
- ☐ D) Hepatolenticular degeneration (Wilson disease)
- ☐ E) Lead poisoning



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

41. An 18-year-old man is brought to the emergency department 1 hour after he fell 25 feet. On examination, he is paraplegic and has urinary incontinence. Examination of the lower extremities shows absent reflexes and flaccid tone. Babinski sign is present bilaterally. X-rays of the spine show a fracture dislocation of T10. Which of the following is the most likely explanation for this patient's urinary incontinence?

- ☐ A) Compression of the conus medullaris
- ☐ B) Compression of the S3–4 nerve roots
- ☐ C) Compression of the T10–11 nerve roots
- ☐ D) Compression of the thoracic spinal cord
- ☐ E) Rupture of the bladder neck



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

42. Four days after admission to the hospital for intravenous antibiotic therapy for a leg rash, a 77-year-old man is unable to urinate. The rash developed 10 days ago on his right foot and has now spread to his calf. On admission, the right lower extremity was warm and erythematous. Yesterday, he had the onset of neck pain. He has type 2 diabetes mellitus currently treated with glyburide. He appears acutely ill. His temperature is 39°C (102.2°F). Examination shows nuchal rigidity. The right lower extremity is warm and erythematous. Muscle strength is full in the proximal upper extremities and is 4/5 in the hands. He is unable to lift the lower extremities against gravity. Deep tendon reflexes are brisk. Babinski sign is present bilaterally. Sensation to pinprick is decreased below C7. Which of the following is the most likely diagnosis?

- ☐ A) Epidural abscess of the spine
- ☐ B) Herniated cervical disc
- ☐ C) Infected hematoma of the spine
- ☐ D) Meningitis
- ☐ E) Subdural empyema



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

43. An otherwise healthy 52-year-old financial advisor comes to the physician because of a 6-week history of progressive right hand weakness that is interfering with his ability to write. There is no history of trauma to the hand. He takes no medications. He is 180 cm (5 ft 11 in) tall and weighs 90 kg (199 lb); BMI is 28 kg/m². Examination of the right hand shows decreased active abduction of the small and index fingers. He cannot cross and uncross his right index and middle fingers, and he is unable to pinch the palmar aspect of the right thumb against the radial side of the index finger without flexing the interphalangeal joint of the thumb to provide strength. Active flexion of the interphalangeal joints of the right hand is 5/5. Sensation over the entire dorsal aspect of the hand is normal. The remainder of the neurologic examination shows no focal findings. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) MRI of the brachial plexus
- ☐ B) MRI of the neck
- ☐ C) Electrophysiological studies
- ☐ D) Biopsy of the sural nerve
- ☐ E) Carpal tunnel release



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

44. A 23-year-old woman with sickle cell disease is brought to the emergency department after becoming obtunded; she has a 2-day history of cough and fever. Her temperature is 39.3°C (102.7°F), pulse is 128/min, respirations are 28/min, and blood pressure is 98/60 mm Hg. She appears stuporous. She is intubated and mechanically ventilated. There is moderate nuchal rigidity. Right basilar crackles are heard on auscultation. Hematocrit is 21% with many nucleated erythrocytes, and leukocyte count is 24,000/mm³. Which of the following is the most appropriate next step in management?
- ☐ A) Administration of antibiotics
 - ☐ B) Administration of dopamine
 - ☐ C) CT scan of the head
 - ☐ D) Transfusion of 3 units of packed red blood cells



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

45. A 57-year-old man comes to the physician because of progressive weakness in both legs over the past 5 months. He has trouble arising from a chair and climbing stairs; he finds it easier to arise from a chair if he rocks back and forth first. During this time, he has also had a persistent cough, a 14-kg (30-lb) weight loss, and a dry mouth. He has smoked two packs of cigarettes daily for 40 years. Examination shows weakness of hip flexion and knee extension. There is no muscle tenderness. All reflexes are absent. Babinski sign is absent. Sensory examination shows no abnormalities. His erythrocyte sedimentation rate is 95 mm/h, and serum creatine kinase activity is 25 U/L. An x-ray of the chest shows a right upper lobe mass. Which of the following is the most likely location of the abnormality?

- ☐ A) Anterior horn cell
- ☐ B) Muscle
- ☐ C) Myoneural junction
- ☐ D) Peripheral nerve axon
- ☐ E) Peripheral nerve myelin
- ☐ F) Spinal cord



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

46. A 52-year-old man who is HIV positive is brought to the physician by his mother because of a 1-month history of forgetfulness and inappropriate behavior. He also has been buying numerous items from the Internet. He also has had diarrhea. During the past year, he has been noncompliant with his antiretroviral medication regimen and has had a 16-kg (35-lb) weight loss. He is 173 cm (5 ft 8 in) tall and weighs 46 kg (101 lb); BMI is 15 kg/m². His temperature is 38.5°C (101.3°F), pulse is 98/min, respirations are 16/min, and blood pressure is 110/60 mm Hg. Examination shows thrush. On mental status examination, he has a labile affect. He recalls zero of three objects after 5 minutes. A CT scan of the head shows prominent atrophy of the cortex and demyelination in the subcortical white matter. Which of the following is the most likely diagnosis?

- ☐ A) Central nervous system lymphoma
- ☐ B) Dementia, Alzheimer type
- ☐ C) Multiple sclerosis
- ☐ D) Pick disease
- ☐ E) Progressive multifocal leukoencephalopathy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

47. A previously healthy 27-year-old woman comes to the physician because of a 1-year history of progressive weakness, numbness, and tingling of both arms. Her pulse is 70/min, respirations are 12/min, and blood pressure is 100/60 mm Hg. Muscle strength is 3/5 in the upper extremities and 4/5 in the lower extremities. There is muscle wasting and atrophy of the forearms and hands bilaterally. Deep tendon reflexes are absent in the upper extremities and 4+ in the lower extremities. Babinski sign is present bilaterally. There is decreased sensation to pinprick and light touch in the upper extremities. Which of the following is the most likely diagnosis?

- ☐ A) Amyotrophic lateral sclerosis
- ☐ B) Guillain-Barré syndrome
- ☐ C) Postherpetic neuralgia
- ☐ D) Syringomyelia
- ☐ E) Tabes dorsalis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

48. A 77-year-old woman is brought to the physician by her husband because of a 3-day history of visual difficulties. He says she seems a little "off balance." She agreed to come in after denting the car on the left side while parking it in the garage. She has not had any headaches or noticed any difficulty using either side of her body. She has not fallen recently. She has a 2-year history of atrial fibrillation treated with aspirin. She is sitting comfortably in a chair. Her temperature is 37°C (98.6°F), pulse is 95/min, respirations are 22/min, and blood pressure is 110/80 mm Hg. Examination shows a left homonymous hemianopia. Muscle strength, muscle tone, and sensation are intact bilaterally. Speech is normal. She is alert and oriented to person, place, and time. When asked to read, she only reads the words on the right side of the page. A CT scan of the head is most likely to show an infarct in the territory of which of the following cerebral arteries?

- ☐ A) Left anterior
- ☐ B) Left middle
- ☐ C) Left posterior
- ☐ D) Right anterior
- ☐ E) Right middle
- ☐ F) Right posterior



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

49. A 72-year-old woman is brought to the physician 2 hours after the sudden onset of dim vision in her right eye. Her symptoms began 3 weeks ago with generalized stiffness, aches in the morning, and a bitemporal headache. Examination shows firm, tender temporal arteries bilaterally. Corrected visual acuity is 20/200 in the right eye and 20/40 in the left. Laboratory studies show:

Hematocrit	32%
Leukocyte count	6000/mm ³
Platelet count	300,000/mm ³
Erythrocyte sedimentation rate	110 mm/h

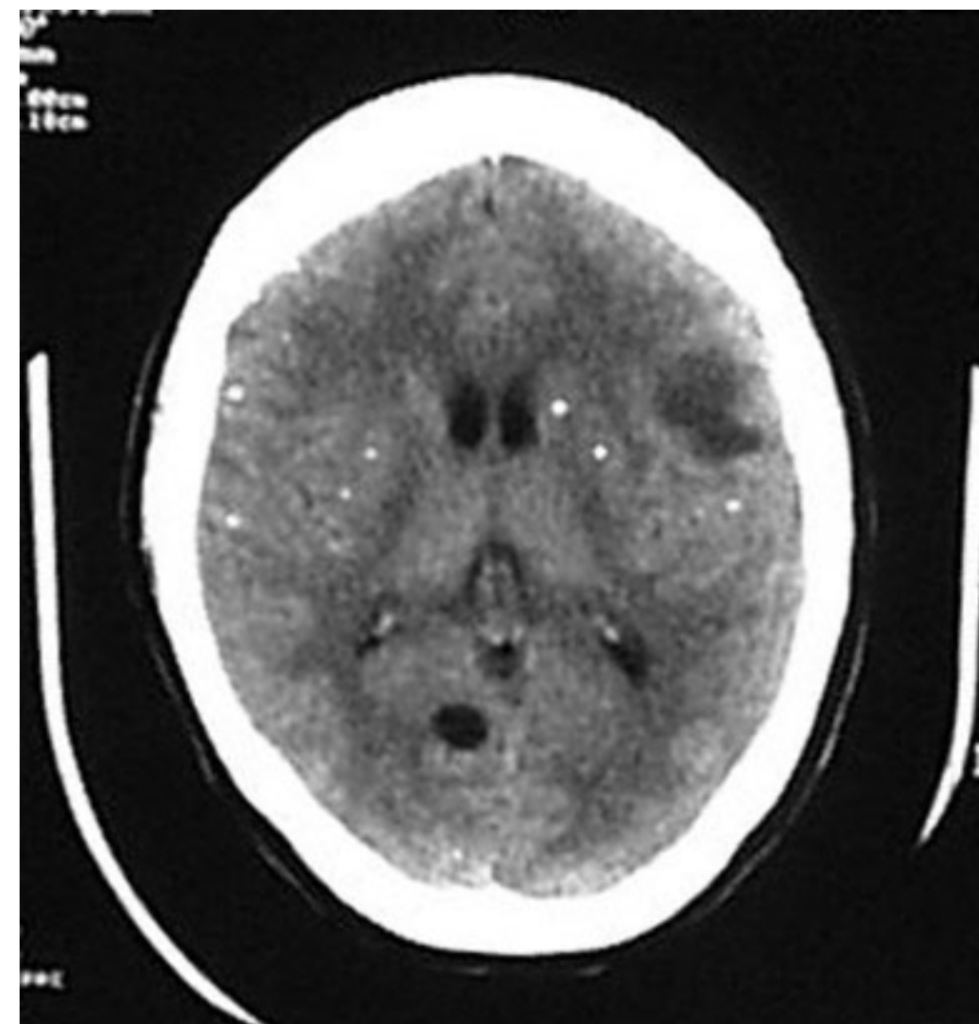
Which of the following is the most appropriate initial step in management?

- ☐ A) Carotid angiography now
- ☐ B) Aspirin therapy now and temporal artery biopsy within 24 hours
- ☐ C) Corticosteroid therapy now and temporal artery biopsy within 3 days
- ☐ D) Plasmapheresis now
- ☐ E) Temporal artery biopsy now with therapy determined by pathology



50. A previously healthy 24-year-old woman is brought to the emergency department by her brother 1 hour after a generalized tonic-clonic seizure that lasted 5 minutes. The patient and her brother emigrated from Central America 6 months ago. She takes no medications. She is drowsy but cooperative. Her temperature is 37.1°C (98.8°F), pulse is 80/min, respirations are 18/min, and blood pressure is 150/90 mm Hg. Examination and laboratory studies show no other abnormalities. Toxicology screening is negative. A CT scan of the head with contrast is shown. Which of the following is the most likely diagnosis?

- ☐ A) Behçet syndrome
- ☐ B) Central nervous system lymphoma
- ☐ C) Metastatic carcinoma
- ☐ D) Neurocysticercosis
- ☐ E) Progressive multifocal leukoencephalopathy
- ☐ F) Toxoplasmosis
- ☐ G) Tuberculous meningitis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause